

HERNAN “ERNIE” MALDONADO

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Summary

Data engineer and ML specialist with 10+ years designing and operating large-scale data pipelines (Hadoop, Spark, Hive, Oozie, SQL) and the production ML systems they feed, across fraud detection, sanctions compliance, and cybersecurity at major financial institutions. Track record building ingestion and ETL pipelines processing hundreds of TBs, cutting processing time by up to 90%, and shipping NLP and gradient-boosted models into production. Holder of three graduate degrees (computational mathematics, public policy, economics) and four ML/deep-learning certifications. Targeting data engineering, ML engineering, or generative AI roles.

Work Experience

Internal Fraud Engineer

Sep 2024 – Present

Citigroup, Tampa FL

Hybrid

- Mine Hadoop databases (transactions, user behavior, historical cases) to surface fraud patterns and translate findings into **PySpark/Hive** detection rules targeting anomalous velocity, high-risk access, and suspicious vendors.
- Collaborate with analysis teams to simulate, tune, and deploy alert logic; manage alert volume by focusing on recurring MOs that mirror confirmed past crimes.

Consumer Protection Big Data & ML Consultant

Oct 2023 – Dec 2023

Bank of America, Tampa FL

Hybrid

- Audited the entire fraud **Hadoop** ecosystem and produced a data-driven benchmark report comparing current practices against industry standards across performance, reliability, and data quality.

Sanctions Machine Learning Engineer

May 2022 – Oct 2023

Citigroup, Tampa FL

Hybrid

- Automated 500K monthly sanctions alerts by building end-to-end NLP pipelines (Random Forest, Gradient Boosting, XGBoost) with 50+ engineered features to match flagged customers against sanctioned-persons lists.
- Cut manual review volume by 50% by discovering Citi’s internal transactions were being double-screened across two independent systems.
- Authored Model Risk Management documentation ensuring compliance with OCC 2011-12 and SR 11-7; produced architecture study showing a Hadoop migration would reduce processing time by 90%.
- **Stack:** SQL, Pandas, Python, Scikit-Learn, PyTorch, XGBoost, FastAPI

Data Scientist – Fraud Detection ML

May 2021 – Apr 2022

Certegy Payments, Tampa FL

Hybrid

- Doubled check-fraud detection rate at Walmart by developing Random Forest, Gradient Boosting, and Logistic Regression models; raised KS statistic from 0.05–0.3 to 0.6–0.8.
- Architected full ML pipeline: features built in Oracle SQL, models served via AWS API with real-time and batch scoring, routed to the correct model per transaction type.
- **Stack:** SAS Viya, Oracle SQL, Oracle OML, Python, PyTorch, Scikit-Learn, TensorFlow, AWS

Sr. Data Scientist

Feb 2021 – Apr 2021

Nielsen, Tampa FL

Remote

- Maintained and extended Java codebase for national-scale TV viewership adjudication; developed new attribution features for programming and advertising data. **Stack:** Java, S3, Git

Cybersecurity Big Data & ML Engineer

Apr 2018 – Feb 2021

Citigroup, Tampa FL

Hybrid

- Built from scratch a production anomaly-detection system monitoring 300K employees across 5TB of weekly data using peer-normalized behavioral metrics and k-means clustering to surface insider threats for the Cyber Security Fusion Center.
- Engineered ingestion pipelines for 12+ data sources into a 270TB Hadoop cluster (Sqoop, Flume, Python parsers, Hive, Oozie) and built a monitoring system to ensure all feed flows remained current.
- **Stack:** PySpark, PyTorch, Scikit-Learn, TensorFlow, Hive, Oozie, Hadoop, Python, SQL, Linux

Data Scientist – Machine Learning

May 2016 – Oct 2017

Wellcare Health Plans (Centene), Tampa FL

On-Site

- Reduced hospital readmissions by 60% using XGBoost and Scikit-Learn, directly limiting Medicare penalty exposure for the plan.
- Key contributor to a \$1M/year capital project coordinating Informatica, Hadoop, SAS, and Python workflows across external vendors and multiple business units.
- **Stack:** PySpark, Hive, Python, R, SQL, XGBoost, Scikit-Learn

Sr. Decision Support Analyst

Jan 2015 – Apr 2016

Gateway HealthPlan (Highmark), Pittsburgh PA

On-Site

- Worked directly with the CFO to resolve reconciliation gaps between insurance claims and accounting systems via ETL improvements; built ARIMA cash-flow forecasting model in R; managed a team of three data engineers and supervised SAS platform administration.

Database & ETL Analyst

Aug 2012 – Oct 2014

UPMC Health Plan, Pittsburgh PA

On-Site

- Built enterprise-wide ETL reporting layers in Oracle SQL and SAS; developed a Random Forest model to segment high-cost members. Earned the 2014 ACES Award (top 1% of UPMC employees) for excellence in service and process improvement.

Project Manager

Jan 2011 – Aug 2012

University of Pittsburgh, Pittsburgh PA

On-Site

- Managed two \$1.5M grant-funded research projects end-to-end (20+ staff), completed data collection ahead of schedule and under budget; projects featured on local television.

Projects

NameGen: Generative AI Name Generator | Python, FastAPI, Docker

[Source Code](#)

- Generates novel names using directed-graph letter-transition probabilities sampled via multinomial distribution – the same statistical foundation underlying language model sampling.
- Deployed three ways: interactive CLI (with pickle-cached graph for speed), local FastAPI service, and Dockerized container, demonstrating a complete ML deployment lifecycle.

E-Commerce Fraud Detection | Python, XGBoost, Optuna, joblib

[Source Code](#)

- Three-model ensemble (Random Forest, Isolation Forest, XGBoost) applied to e-commerce fraud; XGBoost hyperparameters tuned with Optuna.
- Models serialized with joblib pipelines, ready for production deployment.

Education

Duquesne University, Pittsburgh PA USA

2008 - 2011

Masters in Computational Mathematics

3.9 GPA

KDI School, Seoul (Relocated to Sejong City), S. Korea

2004 - 2005

Masters in Public Policy

3.9 GPA

Universidad de los Andes, Bogota, Colombia

2001 - 2003

Masters in Economics

3.7 GPA

Universidad de los Andes, Bogota, Colombia

1996 - 2001

Bachelors in Economics

3.7 GPA

Certificates

AWS Certified Cloud Practitioner

2022

Tensor Flow Specialization

2020

Deep Learning Specialization

2018

Machine Learning Specialization

2017

Machine Learning Certificate

2015

Technical Skills

Languages: Python, SQL, Java, R, SAS, Rust, Matlab

Big Data & Pipelines: Apache Hadoop, Hive, PySpark, Oozie, Impala, Sqoop, Flume

Cloud & Databases: AWS, Oracle DB, MS SQL Server, MySQL

Machine Learning: Scikit-Learn, PyTorch, TensorFlow, XGBoost, Optuna, SAS Viya, PySpark MLlib

GenAI & NLP: LLM APIs, prompt engineering, NLP feature engineering, generative modeling

MLOps & Tools: FastAPI, Docker, Flask, Git, GitHub, GitLab, BitBucket, Postman, Jira

Selected Publications

Participatory Assessment of the Health of Latino Immigrant Men in a Community

2013

Journal of Immigrant and Minority Health

Bayesian Regression Inference Using a Normal Mixture Model

2012

Masters Thesis, Duquesne University

FARC Terrorism in Colombia: A Clustering Analysis

2004

CEDE Working Papers Series